

Project Proposal

Salifort Motors Employee Turnover Prediction

Overview

The objective of this project is to build a predictive machine learning model to identify employees at risk of leaving Salifort Motors. By identifying the key drivers of employee turnover, the HR department can proactively intervene to retain highly skilled talent in a competitive market.

Milestones

- **Plan:** Understand stakeholder needs, perform initial data inspection, and handle data cleaning (duplicates, typos).
- **Analyze:** Conduct Exploratory Data Analysis (EDA) to visualize relationships, specifically focusing on extreme edges of the data where turnover is concentrated.
- **Construct:** Build and evaluate ensemble machine learning models (Random Forest and XGBoost) using all available features.
- **Execute:** Select the Champion Model (XGBoost), extract feature importance, and present actionable recommendations like the "Employee Portfolio" system to leadership.

PACE: Plan Stage

Who are your stakeholders for this project?

The primary stakeholders are the HR department and the Senior Executive Leadership team. HR needs this to retain high-value employees, while Executive Leadership must approve the ethical use of sensitive employee data.

What are you trying to solve or accomplish?

I am building a proactive model to detect which skilled employees are likely to leave. This shifts the company's approach from reactive to predictive, allowing HR to build relationships and retain talent before they seek external opportunities.

What are your initial presumptions about the data that can inform your EDA?

The data is relatively clean with no missing values. The duplicates were removed to prevent bias. I presumed that outliers in `time_spend_company` (tenure) were not errors but represented loyal veterans, so I intentionally kept them for the model to learn from.

Do you have any ethical considerations in this stage?

Yes. Since we are dealing with sensitive HR data, every analytical step and data transformation must be transparently communicated to HR and Executive Leadership. This prevents data leaks and ensures all actions have official approval.

PACE: Analyze Stage

What did you observe about the relationships between variables?

The most crucial observation is that turnover happens at the "extreme edges" of the distributions. Employees with very few projects or an excessive number of projects tend to leave, while the middle interval remains stable. Similarly, early-tenure employees leave for flexibility, and highly tenured employees leave because they have become more skilled and valuable to competitors.

What transformations did you make with your data? Why did you chose to make those decisions?

I standardized column names to snake_case and fixed a crucial typo (average_monthly_hours to average_monthly_hours) to prevent coding errors. I dropped duplicates but retained outliers because tree-based models handle them well, and these outliers represent real, valuable business cases.

What are some purposes of EDA before constructing a predictive model?

EDA helps us visualize systematic patterns and relationships between independent variables and the target variable (left). It confirms that our features contain strong predictive signals rather than just "junk" data.

PACE: Construct Stage

Which independent variables did you choose for the model and why?

I chose to include all independent variables. Because I am using ensemble tree-based models (Random Forest and XGBoost), I did not need to manually drop features. I allowed the algorithms to handle feature selection natively by detecting the optimal splits during training.

How well does your model fit the data?

Both models fit the data exceptionally well, achieving an F1-score and Accuracy of ~99%. XGBoost had a slight edge in recalling actual churners (0.93 vs 0.92).

Can you improve it? Is there anything you would change about the model?

To improve future iterations, I recommend moving away from point-in-time data and creating an "Employee Portfolio" tracking system. This would allow the model to track an employee's skill acquisition and performance growth over 6-month intervals, providing dynamic data on how valuable they are becoming over time.

PACE: Execute Stage

What key insights emerged from your model(s)?

XGBoost was chosen as the Champion Model over Random Forest. While both performed spectacularly, XGBoost is less prone to overfitting on future, unseen data and processes large datasets faster—a critical requirement for handling up-to-date business data. The top drivers for turnover are strictly operational: `average_monthly_hours`, `satisfaction_level`, and `last_evaluation`.

What business recommendations do you propose based on the models built?

- Initiate a "Pilot Testing" phase with HR to test the XGBoost model on incoming new hires and current employees to validate its real-world accuracy without disrupting current operations.
- Implement the "Employee Portfolio" system to continuously track growth and foster a sense of belonging among high-value talent before they transition to competitors.